

POLY MATH INITIATIVE

RESEARCH PROTOCOL

*Memory Management Strategies and Information-Retention Accuracy in Small Open-Weight Language
Models ($\leq 14B$ Parameters)*

1 Research Question & Scope

1.1 Primary Research Question

For open-weight language models $\leq 14B$ parameters, how do different memory architectures — varying in what they store and how they structure and retrieve it — affect factual retention and downstream task accuracy across extended multi-session conversations, and how does this interact with model size within the $\leq 14B$ range?

1.2 Scope

- Model scale: open-weight, instruction-tuned models from 1B to 14B parameters.
- Conversation regime: multi-session dialogue, roughly 6K–100K cumulative tokens (medium-length regime — see §5.4).
- Memory mechanisms in scope: context truncation, summarization, retrieval-augmented memory, and MemGPT-style tiered/paged memory, and combinations thereof (see §3).
- Out of scope: ultra-long-context regimes (1M+ tokens, e.g. BEAM-scale), proprietary/closed-weight models, and fine-tuning of the base models.

2 Background & Motivation

Long-term conversational memory for LLM agents has been studied extensively at the level of individual mechanisms — recursive summarization, retrieval-augmented pipelines, and OS-inspired tiered memory (MemGPT) — and at the level of benchmarks (LoCoMo, LongMemEval, MemoryAgentBench). However, the existing literature overwhelmingly evaluates these mechanisms on a single fixed backbone, most often a large proprietary model or an 7–13B open-weight model used as a fixed default, rather than systematically varying model scale as an independent factor. Where smaller backbones are tested at all, it is typically as a single robustness check (e.g. one 8B ablation) rather than a scaling study.

Separately, position-bias research (“lost in the middle” and related work) has shown that smaller open-weight models are more susceptible to positional and distance-based retrieval bias than larger models, which suggests that memory-strategy rankings established on large backbones may not transfer to the $\leq 14B$ regime. This protocol is designed to close that specific gap: it treats memory architecture and model scale as two crossed independent variables rather than confounding them.

3 Conceptual Framework

3.1 Terminology Note

“Paged memory” is used in this protocol exclusively to mean **MemGPT-style tiered conversational memory** (a working context, a searchable recall store, and an archival store, with movement between tiers governed by explicit function calls). It is distinct from PagedAttention, which manages KV-cache memory at the inference-serving layer and does not itself decide what conversational content is retained, summarized, or forgotten. This protocol does not evaluate PagedAttention or any other inference-serving optimization.

3.2 Two-Axis Design

Rather than treating truncation, summarization, retrieval, and tiered memory as four parallel, mutually exclusive strategies, this protocol decomposes memory architecture into two crossed axes. Summarization, for instance, is treated as a storage-axis choice that can be combined with any access-axis mechanism (including tiered memory, of which it is often a native component).

Axis A — What is stored	Description
Raw turns	Verbatim conversation history, no compression.
Summaries	LLM-generated summaries, single-pass or recursive.
Extracted facts	Structured facts / triples pulled from dialogue.
Graph nodes	Entities and relations linked into a memory graph.

Axis B — How it is structured / accessed	Description
Truncated / sliding window	Oldest content dropped once a length budget is hit.
Full long-context	Entire history force-fit into the model's context window.
Vector retrieval (RAG)	Top-k similarity search over an embedded store.
Tiered / paged (MemGPT-style)	Working set + recall store + archive, moved via explicit calls.
Graph traversal	Entity/relation-based recall over a memory graph.

4 Hypotheses & Sub-Questions

- **H1** — Holding conversation length and metric fixed, the ranking of storage types (raw / summary / fact / graph) for single-hop factual recall differs between the 1–3B and 7–9B size tiers.
- **H2** — Retrieval-based access (RAG) loses proportionally more accuracy to positional / lost-in-the-middle effects at ≤ 14 B than published results show at larger scale, independent of what is stored.
- **H3** — When the small model itself performs memory management (extraction, summarization, relevance judgment) rather than only answering, a meaningful share of the performance gap between architectures is explained by the model's management capability rather than the architecture itself.
- **H4** — There exists a conversation-length threshold below which tiered/paged memory provides no measurable advantage over simple retrieval for models in the 7–14B range.

5 Experimental Design

5.1 Fixed Variables

Variable	Locked value	Rationale
Division of labor	Small model performs both memory management (extraction / summarization /	Tests the harder, more realistic case that most prior work skips by offloading memory management to a larger model.

Variable	Locked value	Rationale
	retrieval judgment) and final answering	
Conversation-length regime	Medium: ~6K–100K cumulative tokens, multi-session	Matches LongMemEval-S and LoCoMo natively; keeps truncation a live, meaningful choice without requiring 1M+-token infrastructure.
Primary metric	Exact-match / F1 against gold answers	Reproducible, cheap, comparable across runs and to published baselines.
Secondary metric	LLM-as-judge correctness score (fixed judge model, fixed rubric)	Captures semantically correct answers that exact-match penalizes; logged separately, never blended into the primary score.
Cost accounting	Tokens processed per query (memory-write + memory-read + generation)	Makes accuracy-per-cost trade-offs visible across architectures.

5.2 Model Matrix

Primary family: Qwen2.5-Instruct. Held constant tokenizer, training recipe, and instruction-tuning pipeline across sizes so that observed differences are attributable to scale rather than data/architecture confounds.

Role	Model	Params	Purpose
Primary — small	Qwen2.5-1.5B-Instruct	1.5B	Tests the weak memory-manager regime.
Primary — mid	Qwen2.5-7B-Instruct	7B	Most-studied size in prior literature; direct comparison to published numbers.
Primary — large	Qwen2.5-14B-Instruct	14B	Upper bound of scope ceiling.

Secondary family (cross-family validation only, not used for the scaling analysis): Llama-3.1/3.2-Instruct.

Role	Model	Params	Purpose
Secondary — small	Llama-3.2-3B-Instruct	3B	Cross-family robustness check.
Secondary — mid	Llama-3.1-8B-Instruct	8B	Most-used backbone in published memory-systems baselines (direct external comparison).

5.3 Axis Combinations Under Test

Full crossing of Axis A $\times$ Axis B is not run for every model; the subset below is prioritized to answer H1–H4 within a bounded compute budget.

#	Storage (Axis A)	Access (Axis B)	Priority
1	Raw turns	Truncated / sliding window	Baseline (required, all models)
2	Raw turns	Full long-context (fits budget)	Baseline (required, all models)

#	Storage (Axis A)	Access (Axis B)	Priority
3	Summaries	Vector retrieval (RAG)	Core comparison (required, all models)
4	Extracted facts	Vector retrieval (RAG)	Core comparison (required, all models)
5	Summaries + raw	Tiered / paged (MemGPT-style)	Core comparison (required, all models)
6	Graph nodes	Graph traversal	Secondary (primary family only, time-permitting)

5.4 Datasets

- **LongMemEval-S** — single/multi-session questions with knowledge-update and temporal-reasoning categories; used for the medium-length regime and for direct comparison to published memory-system baselines.
- **LoCoMo** — very long-term, multi-session persona dialogues with single-hop, multi-hop, open-domain, and temporal-reasoning question categories. Gold-answer noise has been documented in prior work (§7); flagged items should be spot-checked before being treated as ground truth.

6 Metrics

Two metrics are logged for every run; neither is discarded in favor of the other.

- **Primary — Exact-match / F1:** token-level overlap between model answer and gold answer, computed per question category and averaged.
- **Secondary — LLM-as-judge accuracy:** a fixed judge model and fixed rubric score each answer as correct/incorrect against the gold answer, catching semantically correct answers that F1 penalizes for phrasing.
- **Cost — tokens per query:** logged but not used as an accuracy metric; reported alongside accuracy so accuracy-per-cost trade-offs across architectures are visible.

When primary and secondary metrics disagree by more than 10 percentage points for a given cell, that cell is flagged for manual review before being reported.

7 Known Limitations

- **Benchmark noise:** LoCoMo's gold-standard answers have documented inaccuracies and incompleteness in prior published analysis; scores on LoCoMo should be treated as directionally, not absolutely, comparable.
- **Confounded capability vs. architecture:** because the small model performs its own memory management (§5.1), a weak result for a given architecture may reflect the model's limited extraction/summarization ability rather than a flaw in the architecture itself. H3 exists to partially disentangle this but cannot fully separate it.
- **Judge dependency:** the secondary metric inherits any bias or inconsistency of the fixed judge model; judge prompts and outputs should be versioned and spot-audited.
- **Family generalization:** the scaling analysis (H1, H2, H4) is drawn only from the Qwen2.5 family; the Llama secondary family checks direction of effect at one size pair (3B/8B) but does not itself support a scaling claim.

- **Compute and scope ceiling:** the axis-combination subset in §5.3 is prioritized rather than exhaustive; graph-traversal results are secondary and may be underpowered.
- **Regime boundary:** findings are scoped to the medium-length conversational regime (§1.2, §5.1) and should not be extrapolated to ultra-long-context (1M+ token) settings without separate validation.

8 Reference List

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End of protocol.